

Type A - 4 (Out of 7) $\times 10$ pts

Q1. Let V be a finite-dimensional vector space and let T be a linear operator on V . Suppose that $\text{rank}(T^2) = \text{rank}(T)$. Prove that the range and null space of T are disjoint, i.e., have only the zero vector 0 in common.

Hint: You can use the rank-nullity without proof, if needed.

Answer: It is given that $T : V \rightarrow V$ and let $v \in \text{Range}(T) \cap \text{Null}(T)$. Since $v \in \text{Range}(T)$, there exists $u \in V$ such that $T(u) = v$. Also, since $v \in \text{Null}(T)$,

$$T(v) = T(T(u)) = T^2(u) = 0.$$

Hence, $u \in \text{Null}(T^2)$. By the rank-nullity theorem,

$$\text{nullity}(T) = \dim V - \text{rank}(T) \tag{1}$$

and

$$\text{nullity}(T^2) = \dim V - \text{rank}(T^2). \tag{2}$$

Since $\text{rank}(T) = \text{rank}(T^2)$, from (1) and (2) we get

$$\text{nullity}(T) = \text{nullity}(T^2). \tag{3}$$

Now let $w \in \text{Null}(T)$. Then

$$T(w) = 0 \implies T^2(w) = T(T(w)) = 0.$$

This implies $w \in \text{Null}(T^2)$. So,

$$\text{Null}(T) \subseteq \text{Null}(T^2).$$

Using (3), both null spaces have the same dimension, hence

$$\text{Null}(T) = \text{Null}(T^2).$$

Since $u \in \text{Null}(T) = \text{Null}(T^2)$, and we have $T(u) = 0$. Therefore, $v = T(u) = 0$ and this is true for all $v \in \text{Range}(T) \cap \text{Null}(T)$. Hence,

$$\text{Range}(T) \cap \text{Null}(T) = \{0\}.$$

Rubric:

- **Setup (2 Points):** Correctly identifies an element v in the intersection and the existence of its pre-image u .
- **Rank-Nullity (2 Points):** Properly applies the theorem to establish that $\text{nullity}(T) = \text{nullity}(T^2)$.
- **Containment (2 Points):** Formally proves that $\text{Null}(T) \subseteq \text{Null}(T^2)$.
- **Equivalence (2 Points):** Uses the dimension argument to conclude the two null spaces are identical.
- **Conclusion (2 Points):** Successfully shows that v must be the zero vector.

Q2. Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is normal, i.e., $TT^* = T^*T$, and $T(1, 1, 1) = (2, 2, 2)$. Suppose $(z_1, z_2, z_3) \in \text{null } T$ (the null space of T). Prove that $z_1 + z_2 + z_3 = 0$.

Hint: You may use the facts at the end, without proof.

Answer: Let $v = (z_1, z_2, z_3) \in \text{null } T$. As $T \in \mathcal{L}(\mathbb{C}^3)$ is normal, then $v = (z_1, z_2, z_3) \in \text{null } T^*$. Hence $T^*(z_1, z_2, z_3) = 0$. Therefore,

$$\begin{aligned} 2(z_1 + z_2 + z_3) &= \langle (2, 2, 2), (z_1, z_2, z_3) \rangle \\ &= \langle T(1, 1, 1), (z_1, z_2, z_3) \rangle \\ &= \langle (1, 1, 1), T^*(z_1, z_2, z_3) \rangle \\ &= \langle (1, 1, 1), (0, 0, 0) \rangle \\ &= 0. \end{aligned}$$

Therefore, $z_1 + z_2 + z_3 = 0$.

Rubric:

- **Normal Operator Property (3 Points):** Correctly identifies and states that because T is normal, $\text{null}(T) = \text{null}(T^*)$, successfully establishing that $T^*(z_1, z_2, z_3) = 0$.
- **Adjoint Inner Product Application (3 Points):** Translates the problem into an inner product framework and correctly applies the definition of the adjoint operator to move T to T^* (i.e., $\langle Tv, w \rangle = \langle v, T^*w \rangle$).
- **Evaluation of Null Space (2 Points):** Correctly substitutes the null space result to evaluate the inner product as $\langle (1, 1, 1), (0, 0, 0) \rangle = 0$.
- **Algebra & Final Conclusion (2 Points):** Successfully sets the initial expansion of the inner product equal to 0 to conclude $z_1 + z_2 + z_3 = 0$.

Q3. Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by

$$Tv = \langle v, u \rangle x, \text{ for every } v \in V.$$

Suppose the underlying field is $\mathbb{F} = \mathbb{R}$. Prove that T is self-adjoint if and only if u and x are linearly dependent. Does this result remain true if $\mathbb{F} = \mathbb{C}$? Justify.

Answer:

Answer:

Let $u, x \in V$ and define $T \in \mathcal{L}(V)$ by

$$Tv = \langle v, u \rangle x, \quad \forall v \in V.$$

Case 1: $\mathbb{F} = \mathbb{R}$.

$(\Rightarrow)$ Suppose T is self-adjoint. Then for all $v_1, v_2 \in V$,

$$\langle Tv_1, v_2 \rangle = \langle v_1, Tv_2 \rangle.$$

Substituting the definition of T , we obtain

$$\langle \langle v_1, u \rangle x, v_2 \rangle = \langle v_1, \langle v_2, u \rangle x \rangle,$$

which simplifies (by linearity of the inner product) to

$$\langle v_1, u \rangle \langle x, v_2 \rangle = \langle v_2, u \rangle \langle v_1, x \rangle \quad \text{for all } v_1, v_2 \in V.$$

Taking $v_1 = u$ and $v_2 = x$, we get

$$\langle u, u \rangle \langle x, x \rangle = \langle x, u \rangle \langle u, x \rangle = (\langle x, u \rangle)^2.$$

Thus,

$$|\langle x, u \rangle|^2 = \langle u, u \rangle \langle x, x \rangle.$$

By the equality case of the Cauchy-Schwarz inequality, it follows that x and u are linearly dependent.

$(\Leftarrow)$ Conversely, suppose $x = cu$ for some $c \in \mathbb{R}$. Then

$$Tv = \langle v, u \rangle cu = c \langle v, u \rangle u.$$

Hence, for all $v_1, v_2 \in V$,

$$\langle Tv_1, v_2 \rangle = c \langle v_1, u \rangle \langle u, v_2 \rangle,$$

and

$$\langle v_1, Tv_2 \rangle = c \langle v_2, u \rangle \langle v_1, u \rangle.$$

Since $\mathbb{F} = \mathbb{R}$, we have $\langle u, v_2 \rangle = \langle v_2, u \rangle$, and therefore the two expressions are equal. Hence T is self-adjoint.

Case 2: $\mathbb{F} = \mathbb{C}$.

In this case, the inner product satisfies conjugate symmetry:

$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$

Suppose $x = cu$ for some $c \in \mathbb{C}$. Then

$$Tv = c \langle v, u \rangle u.$$

For $v_1, v_2 \in V$, we compute

$$\langle Tv_1, v_2 \rangle = c \langle v_1, u \rangle \langle u, v_2 \rangle = c \langle v_1, u \rangle \overline{\langle v_2, u \rangle},$$

while

$$\langle v_1, Tv_2 \rangle = c \langle v_2, u \rangle \langle v_1, u \rangle.$$

For these to be equal for all $v_1, v_2 \in V$, it is necessary and sufficient that

$$c \langle v_1, u \rangle \overline{\langle v_2, u \rangle} = c \langle v_2, u \rangle \langle v_1, u \rangle \quad \forall v_1, v_2,$$

which holds if and only if $c \in \mathbb{R}$.

So as a conclusion we can say the following:

- Over $\mathbb{R}$, T is self-adjoint if and only if x and u are linearly dependent.
- Over $\mathbb{C}$, T is self-adjoint if and only if $x = cu$ with $c \in \mathbb{R}$.

Rubric:

- **Part (a): $\mathbb{F} = \mathbb{R}$ [6 points]:**

- **2 pts:** Correctly applies the definition of self-adjointness and substitutes $Tv = \langle v, u \rangle x$.
- **2 pts:** Derives the identity

$$\langle v_1, u \rangle \langle x, v_2 \rangle = \langle v_2, u \rangle \langle v_1, x \rangle.$$

- **2 pts:** Correctly concludes x and u are linearly dependent (via Cauchy–Schwarz or equivalent) and proves the converse.

- **Part (b): $\mathbb{F} = \mathbb{C}$ [4 points]:**

- **2 pts:** Recognizes the role of conjugate symmetry in complex inner products.
- **2 pts:** Correctly concludes that T is self-adjoint iff $x = cu$ with $c \in \mathbb{R}$ (or provides an equivalent valid justification).

Q4. (a) Let $n \geq 3$. Let $A = [A_{ij}]$ be an $n \times n$ matrix with entries $A_{ij} = i + j$. Find $\det(A)$.

(b) Let $n \geq 2$. Let $B = [B_{ij}]$ be an $n \times n$ matrix with entries $B_{ij} = ij$. Find $\det(B)$.

Hint: You may use the properties of the determinant instead of the explicit formula.

Answer: (a) Let A be an $n \times n$ matrix with entries $A_{ij} = i + j$, where $n \geq 3$. The first three rows of A are:

- $R_1 = [2, 3, 4, \dots, n+1]$
- $R_2 = [3, 4, 5, \dots, n+2]$
- $R_3 = [4, 5, 6, \dots, n+3]$

By Lemma 2, Property 4, subtracting one row from another leaves the determinant unchanged. Let us perform two row operations:

- (a) Replace R_3 with $R_3 - R_2$. The new third row is $R'_3 = [1, 1, 1, \dots, 1]$.
- (b) Replace R_2 with $R_2 - R_1$. The new second row is $R'_2 = [1, 1, 1, \dots, 1]$.

The resulting matrix has the same determinant as A , but its second and third rows are now identical ($R'_2 = R'_3$). By Lemma 2, Property 1, if any two rows of a matrix are equal, its determinant is 0. Therefore, $\det(A) = 0$.

(b) Let B be an $n \times n$ matrix with entries $B_{ij} = ij$, where $n \geq 2$. The first two rows of B are:

- $R_1 = [1, 2, 3, \dots, n]$
- $R_2 = [2, 4, 6, \dots, 2n]$

By Lemma 2, Property 4, subtracting a multiple of one row from another leaves the determinant unchanged. We can evaluate this in two simple ways:

- *Method 1 (Equal Rows):* Replace R_2 with $R_2 - R_1$. The new second row becomes $R'_2 = [1, 2, 3, \dots, n]$. Now the first and second rows are identical ($R_1 = R'_2$). By Lemma 2, Property 1, the determinant is zero.
- *Method 2 (Zero Row):* Replace R_2 with $R_2 - 2R_1$. The new second row becomes $R'_2 = [0, 0, 0, \dots, 0]$. By Lemma 2, Property 2, if a matrix has a zero row, its determinant is zero.

Therefore, $\det(B) = 0$.

Rubric:

- **Part (a) [5 points]:**
 - * **2 pts:** Correctly states that $\det(A) = 0$.

- * **3 pts:** Provides valid justification using the properties of determinants (e.g., explicitly performing row operations like $R_3 - R_2$ and $R_2 - R_1$ [**2 pts**] to produce two equal rows, and explicitly citing the property that a matrix with two equal rows has a determinant of zero [**1 pt**]).
- **Part (b) [5 points]:**
 - * **2 pts:** Correctly states that $\det(B) = 0$.
 - * **3 pts:** Provides valid justification using the properties of determinants (e.g., performing a row operation like $R_2 - R_1$ to create two equal rows or $R_2 - 2R_1$ to create a zero row [**2 pts**], and explicitly citing the corresponding determinant property [**1 pt**]).

Q5. Consider the operator $T \in \mathcal{L}(\mathbb{F}^5)$ that acts as

$$T(x, y, z, u, v) = (x, y + 2z + 3u + 4v, 3z + 2u + 3v, 4u + 2v, 5v).$$

Prove that there *exists* a $\lambda \in F$ and a 2-dimensional subspace U of $\mathbb{F}^5$ such that

- U is invariant under T and
- for every $u \in U$, $Tu = \lambda u$.

Note that proving existence is enough.

Hint: This question requires almost no computation to answer.

What is the matrix representation of T ? Maybe focus on the first row and column; and then focus on the rest of the matrix.

You can use theorem statements from the text, without proof, in your argument.

Answer: The matrix of T in the standard basis of F^5 is

$$[T]_{\text{std}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 3 & 2 & 3 \\ 0 & 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}.$$

So the eigenvalues are $\{1, 1, 3, 4, 5\}$. For $\lambda = 1$, we solve $(T - I)v = 0$, i.e.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 & 3 \\ 0 & 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

From the system of equations:

$$4v_5 = 0 \Rightarrow v_5 = 0,$$

$$3v_4 + 2v_5 = 0 \Rightarrow v_4 = 0,$$

$$2v_3 + 2v_4 + 3v_5 = 0 \Rightarrow v_3 = 0.$$

Thus v_1, v_2 are free variables. Hence, with respect to the standard basis of F^5 , the eigenspace corresponding to $\lambda = 1$ is

$$E_1 = \{(x, y, 0, 0, 0) \in F^5 : x, y \in F\}.$$

Therefore, $\dim(E_1) = 2$.

A basis of E_1 is $E_1 = \text{span}\{e_1, e_2\}$, where e_1, e_2 are the first two vectors of the standard basis of F^5 .

Finally, if $u = (a, b, 0, 0, 0) \in E_1$, then $T(u) = (a, b, 0, 0, 0) = u$, so $Tu = 1 \cdot u$ for all $u \in E_1$. Therefore, for every $u \in U$, $Tu = \lambda u$ with $\lambda = 1$.

Rubric:

- [**2 pts**]: Correctly identifies $\lambda = 1$ as an eigenvalue (e.g., from the diagonal / structure of the matrix).
- [**2 pts**]: Correctly constructs a 2-dimensional subspace $U = \text{span}\{e_1, e_2\}$ (or equivalent eigenspace).
- [**2 pts**]: Shows that U is invariant under T (e.g., $T(e_1) = e_1$, $T(e_2) = e_2$).
- [**2 pts**]: Verifies that for all $u \in U$, $Tu = \lambda u$.
- [**2 pts**]: Clearly concludes existence of such λ and U .

Q6. Let V be finite-dimensional (over the field $\mathbb{F}$) and $\phi : V \rightarrow \mathbb{F}$ be a linear functional on V . Show that there is some vector $u \in V$ such that $\phi(v) = \langle v, u \rangle$ for all $v \in V$ (no need to show uniqueness).

Hint: Start by writing v as a linear combination of standard basis vectors, substitute it in $\phi(v)$ and use the linearity of ϕ .

Answer: Let $e_1, \dots, e_n$ be an orthonormal basis of V . Then we can write

$$v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n.$$

By substituting it we have

$$\begin{aligned} \phi(v) &= \phi(\langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n) \\ &= \langle v, e_1 \rangle \phi(e_1) + \dots + \langle v, e_n \rangle \phi(e_n) \quad [\text{Using linearity of } \phi] \\ &= \left\langle v, \overline{\phi(e_1)} e_1 + \dots + \overline{\phi(e_n)} e_n \right\rangle \end{aligned}$$

for every $v \in V$. Thus setting

$$u = \overline{\phi(e_1)} e_1 + \dots + \overline{\phi(e_n)} e_n,$$

we have $\phi(v) = \langle v, u \rangle$ for every $v \in V$.

Rubric:

- **Basis Selection (2 Points):** Correctly establishes and uses an *orthonormal* basis $e_1, \dots, e_n$ for V . (Using just any basis without orthogonal projection formulas makes the construction much harder).
- **Vector Expansion (2 Points):** Accurately writes an arbitrary vector v as a linear combination of the orthonormal basis elements using inner products: $v = \sum \langle v, e_i \rangle e_i$.
- **Linearity Application (2 Points):** Correctly applies the linearity of the functional ϕ to distribute it across the sum and pull out the inner product scalar coefficients.
- **Inner Product Properties (2 Points):** Correctly brings the scalars $\phi(e_i)$ inside the second argument of the inner product by taking their complex conjugates $\overline{\phi(e_i)}$.
- **Conclusion / Construction of u (2 Points):** Successfully extracts the second argument of the consolidated inner product to define the final vector $u = \sum \overline{\phi(e_i)} e_i$ and concludes the proof.

Q7. Suppose $0 < \theta < \pi/2$. Demonstrate the Gram-Schmidt procedure on the following vectors (in $\mathbb{R}^2$)

$$(\cos \theta, \sin \theta) \text{ and } (0, 1)$$

by proceeding in two different ways:

- Treat $(0, 1)$ as the first vector.**
- Treat $(\cos \theta, \sin \theta)$ as the first vector.**

By the first vector, it is meant that this vector should be part of the final orthonormal basis.

Answer:

- Let $v_1 = (0, 1)$. Hence $\|v_1\| = 1$. Therefore,

$$e_1 = \frac{v_1}{\|v_1\|} = (0, 1).$$

Now, $v_2 = (\cos \theta, \sin \theta)$. This suggests $\langle v_2, e_1 \rangle = \langle (\cos \theta, \sin \theta), (0, 1) \rangle = \sin \theta$.

Also, $v_2 - \langle v_2, e_1 \rangle e_1 = (\cos \theta, \sin \theta) - (0, \sin \theta) = (\cos \theta, 0)$. Therefore,

$$e_2 = \frac{v_2 - \langle v_2, e_1 \rangle e_1}{\|v_2 - \langle v_2, e_1 \rangle e_1\|} = \frac{(\cos \theta, 0)}{\cos \theta} = (1, 0).$$

Hence, $e_1 = (0, 1)$ and $e_2 = (1, 0)$.

- Let $v_1 = (\cos \theta, \sin \theta)$. Hence $\|v_1\| = 1$. Therefore,

$$e_1 = \frac{v_1}{\|v_1\|} = (\cos \theta, \sin \theta).$$

Now, $v_2 = (0, 1)$. This suggests $\langle v_2, e_1 \rangle = \langle (0, 1), (\cos \theta, \sin \theta) \rangle = \sin \theta$.

Also, $v_2 - \langle v_2, e_1 \rangle e_1 = (0, 1) - \sin \theta (\cos \theta, \sin \theta) = (-\sin \theta \cos \theta, \cos^2 \theta) = \cos \theta (-\sin \theta, \cos \theta)$. Therefore,

$$e_2 = \frac{v_2 - \langle v_2, e_1 \rangle e_1}{\|v_2 - \langle v_2, e_1 \rangle e_1\|} = \frac{\cos \theta (-\sin \theta, \cos \theta)}{\cos \theta} = (-\sin \theta, \cos \theta).$$

Hence, $e_1 = (\cos \theta, \sin \theta)$ and $e_2 = (-\sin \theta, \cos \theta)$.

Rubric:

- (a) [1 pt] Correctly identifies $e_1 = (0, 1)$ with justification that $\|v_1\| = 1$.
[1 pt] Correctly computes $\langle v_2, e_1 \rangle = \sin \theta$.
[1 pt] Correctly computes the subtraction $v_2 - \langle v_2, e_1 \rangle e_1 = (\cos \theta, 0)$.
[1 pt] Correctly computes the norm $\|(\cos \theta, 0)\| = \cos \theta$.
[1 pt] Correctly normalizes to obtain $e_2 = (1, 0)$.
- (b) [1 pt] Correctly identifies $e_1 = (\cos \theta, \sin \theta)$ with justification that $\|v_1\| = 1$.
[1 pt] Correctly computes $\langle v_2, e_1 \rangle = \sin \theta$.
[1 pt] Correctly computes the subtraction $v_2 - \langle v_2, e_1 \rangle e_1 = \cos \theta(-\sin \theta, \cos \theta)$.
[1 pt] Correctly computes the norm $\|\cos \theta(-\sin \theta, \cos \theta)\| = \cos \theta$.
[1 pt] Correctly normalizes to obtain $e_2 = (-\sin \theta, \cos \theta)$.
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Type B - 3 (Out of 4) × 16 pts

Q8. Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

- Find matrix of T with respect to the ordered standard basis $\{e_1, e_2, e_3\}$.
- Obtain the row reduced echelon form of the matrix of T (in the standard basis). Show the calculation explicitly.
- Using the calculation in (b), find the matrix for T^{-1} (in the standard basis).

Answer:

Let $T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$

- Matrix of T in the standard basis**

Compute:

$$T(e_1) = (3, -2, -1), \quad T(e_2) = (0, 1, 2), \quad T(e_3) = (1, 0, 4).$$

Thus,

$$A = [T] = \begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}.$$

- Row-reduced echelon form**

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Hence,

$$\text{RREF}(A) = I_3,$$

So T is invertible.

- Matrix of T^{-1}**

Augment and row-reduce:

$$\left[\begin{array}{ccc|ccc} 3 & 0 & 1 & 1 & 0 & 0 \\ -2 & 1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{4}{9} & \frac{2}{9} & -\frac{1}{9} \\ 0 & 1 & 0 & \frac{13}{9} & \frac{2}{9} & -\frac{2}{9} \\ 0 & 0 & 1 & -\frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{array} \right].$$

Thus,

$$T^{-1} = \begin{bmatrix} \frac{4}{9} & \frac{2}{9} & -\frac{1}{9} \\ \frac{13}{9} & \frac{2}{9} & -\frac{2}{9} \\ -\frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}.$$

Rubric:

Part (a): 5 marks

- Correctly identifies the images of the basis vectors $T(e_1), T(e_2), T(e_3)$: **2 marks**
- Writes the three images correctly in coordinate form: **1 mark**
- Forms the matrix of T with correct arrangement of coefficients/columns: **2 marks**

Part (b): 6 marks

- Sets up the correct matrix of T for row reduction: **1 mark**
- Shows a valid first row operation or normalization step correctly: **1 mark**
- Produces a correct intermediate echelon form after eliminating entries below the first pivot: **2 marks**
- Completes the elimination to obtain pivots in the second and third columns: **1 mark**
- Reaches the final RREF I_3 : **1 mark**

Part (c): 5 marks

- Correctly augments the matrix with the identity matrix: **1 mark**

- Applies row operations consistently from part (b) or performs equivalent valid reduction: **2 marks**
- Obtains the correct reduced augmented matrix with I_3 on the left: **1 mark**
- Extracts the inverse matrix correctly from the right-hand side: **1 mark**

Q9. Suppose $T \in \mathcal{L}(V)$ has a singular value decomposition given by

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_n \langle v, e_n \rangle f_n$$

for every $v \in V$, where $s_1, s_2, \dots, s_n$ are strictly positive singular values of T and $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are orthonormal bases of V .

(a) Prove that if $v \in V$, then

$$T^*v = s_1 \langle v, f_1 \rangle e_1 + \cdots + s_n \langle v, f_n \rangle e_n.$$

(b) Prove that if $v \in V$, then

$$T^*Tv = s_1^2 \langle v, e_1 \rangle e_1 + \cdots + s_n^2 \langle v, e_n \rangle e_n.$$

(c) Suppose $S \in \mathcal{L}(V)$ is defined by, for every $v \in V$,

$$Sv = \frac{\langle v, f_1 \rangle e_1}{s_1} + \cdots + \frac{\langle v, f_n \rangle e_n}{s_n}.$$

Prove that ST is the identity operator on V .

Answer: We have $Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_n \langle v, e_n \rangle f_n = \sum_{k=1}^n s_k \langle v, e_k \rangle f_k$.

(a) Let $w \in V$. Now,

$$\begin{aligned} \langle Tv, w \rangle &= \left\langle \sum_{k=1}^n s_k \langle v, e_k \rangle f_k, w \right\rangle \\ &= \sum_{k=1}^n s_k \langle v, e_k \rangle \langle f_k, w \rangle \\ &= \sum_{k=1}^n s_k \overline{\langle w, f_k \rangle} \langle v, e_k \rangle \\ &= \left\langle v, \sum_{k=1}^n s_k \langle w, f_k \rangle e_k \right\rangle \\ &= \langle v, T^*w \rangle. \end{aligned}$$

Here, we used the fact that singular values are real and the inner product is conjugate-linear in the second argument. Therefore,

$$T^*v = \sum_{k=1}^n s_k \langle v, f_k \rangle e_k = s_1 \langle v, f_1 \rangle e_1 + \cdots + s_n \langle v, f_n \rangle e_n.$$

(b) We have

$$T^*Tv = T^*(Tv) = \sum_{k=1}^n s_k \langle Tv, f_k \rangle e_k.$$

Now,

$$\langle Tv, f_k \rangle = \left\langle \sum_{j=1}^n s_j \langle v, e_j \rangle f_j, f_k \right\rangle = \sum_{j=1}^n s_j \langle v, e_j \rangle \langle f_j, f_k \rangle = \sum_{j=1}^n s_k \langle v, e_j \rangle \delta_{jk} = s_k \langle v, e_k \rangle,$$

where δ_{jk} is Kronecker-Delta and is equal to 1 when $j = k$, otherwise 0. We use this fact as $f_1, \dots, f_n$ is the orthonormal basis of V . Using this,

$$T^*Tv = \sum_{k=1}^n s_k^2 \langle v, e_k \rangle f_k = s_1^2 \langle v, e_1 \rangle e_1 + \cdots + s_n^2 \langle v, e_n \rangle e_n.$$

(c) We have $Sv = \frac{\langle v, f_1 \rangle e_1}{s_1} + \dots + \frac{\langle v, f_n \rangle e_n}{s_n} = \sum_{k=1}^n \frac{\langle v, f_k \rangle e_k}{s_k}$. Now,

$$STv = S(Tv) = \sum_{k=1}^n \frac{\langle Tv, f_k \rangle e_k}{s_k}.$$

Now, we already have

$$\langle Tv, f_k \rangle = s_k \langle v, e_k \rangle.$$

Using this,

$$STv = \sum_{k=1}^n \frac{s_k \langle v, e_k \rangle e_k}{s_k} = \sum_{k=1}^n \langle v, e_k \rangle e_k = v = \mathbb{I}v.$$

This implies $(ST - \mathbb{I})v = 0$. As we know that the operator that sends every vector $v \in V$ to 0 is itself the zero operator, we can conclude

$$ST - \mathbb{I} = 0 \implies ST = \mathbb{I}.$$

Rubric:

- (a) [2 pt] Correctly use $\langle Tv, w \rangle$ and write the expression.
 [4 pt] Correctly shows the calculation $\langle Tv, w \rangle = \langle v, T^*w \rangle$.
 [-0.5 pt] Not showing the conjugate linearity of the inner product.
- (b) [1 pt] Correctly writes $T^*Tv = T^*(Tv)$.
 [2 pt] Correctly expands $\langle Tv, f_k \rangle$ using the SVD formula for Tv .
 [2 pt] Correctly concludes $T^*Tv = \sum_{k=1}^n s_k^2 \langle v, e_k \rangle e_k$.
- (c) [1 pt] Correctly writes $STv = S(Tv)$ and applies the definition of S .
 [1 pt] Correctly invokes $\langle Tv, f_k \rangle = s_k \langle v, e_k \rangle$ from part (b) (or re-derives it).
 [2 pt] Correctly calculates $STv = \mathbb{I}v$.
 [1 pt] Correctly concludes $(ST - \mathbb{I})v = 0$ for all $v \in V$, hence $ST - \mathbf{I} = 0 \implies ST = \mathbf{I}$.

Q10. Suppose $T \in \mathcal{L}(V)$ is a linear operator on the vector space V . Show that for each i , U_i (as described below) is an invariant subspace of V .

- (a) Let T and V be arbitrary and define $U_1 := \text{nullspace}(T)$, $U_2 := \text{range}(T)$. Now suppose T has an eigenvalue λ and define $U_3 := \text{range}(T - \lambda I)$.
Hint: It may help to write Tu in terms of $(T - \lambda I)u$.
- (b) Let V be the vector space of polynomials of degree at most 4. Let $T \in \mathcal{L}(V)$ be the differentiation operator. For each $k \in \{1, 2, 3, 4\}$, let $U_k := \text{set of polynomials of degree at most } k$.
- (c) Let $B := (v_1, \dots, v_n)$ be a basis for V and let $T \in \mathcal{L}(V)$ be an operator such that $[T]_B$ is an upper-triangular matrix. For each $k \in \{1, \dots, n\}$, let $U_k := \text{span}\langle v_1, \dots, v_k \rangle$.

Answer: We prove in each case that $T(U_i) \subseteq U_i$, i.e., each U_i is an invariant subspace.

(a)

(i) $U_1 = \ker(T)$.

Let $u \in U_1$. Then $Tu = 0$. Hence,

$$T(Tu) = T(0) = 0,$$

so $Tu \in \ker(T) = U_1$. Therefore,

$$T(U_1) \subseteq U_1.$$

(ii) $U_2 = \text{range}(T)$.

Let $u \in U_2$. Then $u = Tv$ for some $v \in V$. Hence,

$$Tu = T(Tv) = T^2v.$$

Since $T^2v = T(Tv)$, it follows that $Tu \in \text{range}(T) = U_2$. Thus,

$$T(U_2) \subseteq U_2.$$

(iii) $U_3 = \text{range}(T - \lambda I)$, where λ is an eigenvalue.

Let $u \in U_3$. Then $u = (T - \lambda I)v$ for some $v \in V$. Now,

$$Tu = T((T - \lambda I)v).$$

Write $T = (T - \lambda I) + \lambda I$. Then,

$$Tu = ((T - \lambda I) + \lambda I)(T - \lambda I)v.$$

Expanding,

$$Tu = (T - \lambda I)^2v + \lambda(T - \lambda I)v.$$

Factor:

$$Tu = (T - \lambda I)((T - \lambda I)v + \lambda v).$$

Hence $Tu \in \text{range}(T - \lambda I) = U_3$, so

$$T(U_3) \subseteq U_3.$$

(b)

Let V be the vector space of polynomials of degree at most 4, and $T = \frac{d}{dx}$.

For $k \in \{1, 2, 3, 4\}$, let

$$U_k = \{p(x) \in V : \deg p \leq k\}.$$

Let $p(x) \in U_k$. Then $\deg p \leq k$, so $\deg p' \leq k - 1 \leq k$. Hence $p' \in U_k$. Therefore,

$$T(U_k) \subseteq U_k,$$

so U_k is invariant.

(c)

Let $B = (v_1, \dots, v_n)$ be a basis of V such that $[T]_B$ is upper triangular. For each k , define

$$U_k = \text{span}(v_1, \dots, v_k).$$

Let $1 \leq j \leq k$. Since $[T]_B$ is upper triangular,

$$T(v_j) = \sum_{i=1}^n a_{ij}v_i, \quad \text{with } a_{ij} = 0 \text{ for } i > j.$$

Hence,

$$T(v_j) \in \text{span}(v_1, \dots, v_j) \subseteq \text{span}(v_1, \dots, v_k) = U_k.$$

Therefore,

$$T(U_k) \subseteq U_k,$$

so U_k is invariant.

Conclusion: In all cases, each U_i is invariant under T .

Rubric:

- **Part (a) [8 points]:**
 - (i) **$\ker(T)$ invariant [2 pts]:**
 - * **1 pt:** Takes $u \in \ker(T)$ and uses $Tu = 0$.
 - * **1 pt:** Shows $T(Tu) = 0 \Rightarrow Tu \in \ker(T)$.
 - (ii) **$\text{range}(T)$ invariant [2 pts]:**
 - * **1 pt:** Writes $u = Tv$ for some $v \in V$.
 - * **1 pt:** Shows $Tu = T^2v \in \text{range}(T)$.
 - (iii) **$\text{range}(T - \lambda I)$ invariant [4 pts]:**
 - * **1 pt:** Writes $u = (T - \lambda I)v$.
 - * **2 pts:** Correctly manipulates Tu using $T = (T - \lambda I) + \lambda I$ (or equivalent argument).
 - * **1 pt:** Concludes $Tu \in \text{range}(T - \lambda I)$.
- **Part (b) [4 points]:**
 - **2 pts:** Correctly identifies $T = \frac{d}{dx}$ and $U_k = \{p : \deg p \leq k\}$.
 - **2 pts:** Shows $\deg(p') \leq k$ (or $\leq k - 1$) and concludes $T(U_k) \subseteq U_k$.
- **Part (c) [4 points]:**

- **2 pts:** Uses upper-triangular structure to argue $T(v_j) \in \text{span}(v_1, \dots, v_j)$.
- **2 pts:** Concludes $T(U_k) \subseteq U_k$.

Q11. This question asks you to fill in some steps that go into the proof of the following statement:

Suppose $u, v \in V$ where V is an inner product space. Then,

$$\|u + v\| \leq \|u\| + \|v\| \quad (1)$$

and that equality holds if and only if $\langle u, v \rangle = \|u\| \cdot \|v\|$.

- (a) **Show that** $\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle)$ **where** $\Re(\cdot)$ **denotes the real part.**
 (b) **Show that**

$$\|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle) \leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle| \quad (2)$$

$$\leq \|u\|^2 + \|v\|^2 + 2 \cdot \|u\| \cdot \|v\|. \quad (3)$$

Using your results so far, prove Inequality (1).

- (c) **It is clear that Inequality (1) is saturated (i.e. the inequality becomes an equality), if and only if, both inequalities (2) and (3) saturate.**

Show that this, in turn, is equivalent to $\langle u, v \rangle = \|u\| \cdot \|v\|$.

Hint: You can use this fact (without proof): $|\alpha| = \Re(\alpha)$ *implies* $\alpha = |\alpha|$, *for any complex number* α .

Answer: (a) Show that $\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle)$

Using the definition of the norm squared as the inner product of a vector with itself, we can expand $\|u + v\|^2$:

$$\|u + v\|^2 = \langle u + v, u + v \rangle$$

Using the additivity in the first and second slots of the inner product, we can distribute this expression:

$$\langle u + v, u + v \rangle = \langle u, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \langle v, u \rangle$$

By the definition of the norm, $\langle u, u \rangle = \|u\|^2$ and $\langle v, v \rangle = \|v\|^2$. By the conjugate symmetry property of inner products, $\langle v, u \rangle = \overline{\langle u, v \rangle}$. Therefore, the sum of these two terms is twice the real part of the complex number $\langle u, v \rangle$:

$$\langle u, v \rangle + \overline{\langle u, v \rangle} = 2\Re(\langle u, v \rangle)$$

Substituting these identities back into the expansion yields:

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle)$$

(b) Show Inequalities (2) and (3), and prove Inequality (1)

For any complex number z , the real part is always less than or equal to its absolute value, meaning $\Re(z) \leq |z|$. Applying this to the inner product $\langle u, v \rangle$, we get $\Re(\langle u, v \rangle) \leq |\langle u, v \rangle|$. Substituting this into the result from part (a) proves Inequality (2):

$$\|u\|^2 + \|v\|^2 + 2\Re(\langle u, v \rangle) \leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle| \quad \text{— (2)}$$

Next, by the Cauchy-Schwarz Inequality, we know that the absolute value of the inner product is bounded by the product of the norms: $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Substituting this upper bound proves Inequality (3):

$$\|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle| \leq \|u\|^2 + \|v\|^2 + 2\|u\| \cdot \|v\| \quad \text{— (3)}$$

Chaining these inequalities together, we obtain:

$$\|u + v\|^2 \leq \|u\|^2 + \|v\|^2 + 2\|u\| \cdot \|v\|$$

Notice that the right side is a perfect square:

$$\|u + v\|^2 \leq (\|u\| + \|v\|)^2$$

Taking the positive square root of both sides gives us Inequality (1), the Triangle Inequality:

$$\|u + v\| \leq \|u\| + \|v\| \quad \text{— (1)}$$

(c) Show the equality condition is equivalent to $\langle u, v \rangle = \|u\| \cdot \|v\|$

Inequality (1) is saturated (becomes an equality) if and only if both inequalities (2) and (3) saturate.

- Saturation of (2) implies $\Re(\langle u, v \rangle) = |\langle u, v \rangle|$. Using the provided hint, if the real part of a complex number equals its absolute value, the number must be purely real and non-negative. Therefore, we can conclude that $\langle u, v \rangle = |\langle u, v \rangle|$.
- Saturation of (3) means the Cauchy-Schwarz inequality becomes an equality, which requires $|\langle u, v \rangle| = \|u\| \cdot \|v\|$.

Substituting the first deduction ($\langle u, v \rangle = |\langle u, v \rangle|$) into the second deduction yields the required equivalence:

$$\langle u, v \rangle = \|u\| \cdot \|v\|$$

(Conversely, if $\langle u, v \rangle = \|u\| \cdot \|v\|$, then both $\Re(\langle u, v \rangle)$ and $|\langle u, v \rangle|$ equal $\|u\| \cdot \|v\|$, saturating both inequalities).

Rubric:

- **Part (a) [5 points]:**
 - **2 pts:** Correctly expands the inner product $\langle u + v, u + v \rangle$ into four terms.
 - **2 pts:** Correctly applies conjugate symmetry to show $\langle v, u \rangle = \overline{\langle u, v \rangle}$ and defines norm squared terms.
 - **1 pt:** Concludes that the sum of the complex inner product and its conjugate forms $2\Re(\langle u, v \rangle)$.
 - **Part (b) [6 points]:**
 - **2 pts:** States the property of complex numbers that $\Re(z) \leq |z|$ to justify Inequality (2).
 - **2 pts:** Explicitly cites and correctly applies the Cauchy-Schwarz Inequality to justify Inequality (3).
 - **2 pts:** Correctly factors the resulting expression into $(\|u\| + \|v\|)^2$ and takes the square root to conclude the proof for (1).
 - **Part (c) [5 points]:**
 - **2 pts:** Notes that saturation of (2) means $\Re(\langle u, v \rangle) = |\langle u, v \rangle|$.
 - **1 pt:** Uses the given hint to deduce that $\langle u, v \rangle$ must be a non-negative real number, meaning $\langle u, v \rangle = |\langle u, v \rangle|$.
 - **1 pt:** Notes that saturation of (3) means Cauchy-Schwarz is an equality: $|\langle u, v \rangle| = \|u\| \cdot \|v\|$.
 - **1 pt:** Combines these facts to reach the final equivalence $\langle u, v \rangle = \|u\| \cdot \|v\|$.
-

Q12. Let V and W be vector spaces over the field F and let T be a linear transformation from V to W . Suppose the vectors $\{\alpha_1, \dots, \alpha_k\}$ form a basis for the null space of T . Let $\{\alpha_{k+1}, \dots, \alpha_n\}$ be vectors such that $\{\alpha_1, \dots, \alpha_n\}$ are a basis for V . Show that $\{T\alpha_{k+1}, \dots, T\alpha_n\}$ are linearly independent and span the range of T .

Answer:

To show that the set $\{T\alpha_{k+1}, \dots, T\alpha_n\}$ forms a basis for the range of T , $R(T)$, we must prove two conditions: (1) the set spans $R(T)$, and (2) the set is linearly independent.

Part 1: The set spans $R(T)$.

Let w be an arbitrary vector in $R(T)$. By definition of the range, there exists some vector $v \in V$ such that $T(v) = w$. Since $\{\alpha_1, \dots, \alpha_n\}$ is a basis for V , we can write v as a linear combination of these basis vectors:

$$v = c_1\alpha_1 + \dots + c_k\alpha_k + c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n$$

where $c_i \in F$. Applying the linear transformation T to both sides, we get:

$$w = T(v) = T(c_1\alpha_1 + \dots + c_k\alpha_k + c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n)$$

By the linearity of T , this expands to:

$$w = c_1T(\alpha_1) + \dots + c_kT(\alpha_k) + c_{k+1}T(\alpha_{k+1}) + \dots + c_nT(\alpha_n)$$

We are given that $\{\alpha_1, \dots, \alpha_k\}$ is a basis for the null space $N(T)$. Therefore, $T(\alpha_i) = 0$ for $i = 1, \dots, k$. Substituting these zero vectors into our equation leaves:

$$w = 0 + \dots + 0 + c_{k+1}T(\alpha_{k+1}) + \dots + c_nT(\alpha_n)$$

$$w = c_{k+1}T(\alpha_{k+1}) + \dots + c_nT(\alpha_n)$$

Since w was an arbitrary vector in $R(T)$, this shows that any vector in $R(T)$ can be written as a linear combination of the set $\{T\alpha_{k+1}, \dots, T\alpha_n\}$. Thus, the set spans $R(T)$.

Part 2: The set is linearly independent.

Suppose we have a linear combination of these vectors set to the zero vector:

$$c_{k+1}T(\alpha_{k+1}) + \dots + c_nT(\alpha_n) = 0$$

By the linearity of T , we can combine the left side:

$$T(c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n) = 0$$

Let $u = c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n$. The equation $T(u) = 0$ implies that u is in the null space of T , meaning $u \in N(T)$. Since $\{\alpha_1, \dots, \alpha_k\}$ is a basis for $N(T)$, we can write u as a linear combination of these null space basis vectors:

$$u = b_1\alpha_1 + \dots + b_k\alpha_k$$

for some scalars $b_i \in F$. Equating our two expressions for u :

$$b_1\alpha_1 + \dots + b_k\alpha_k = c_{k+1}\alpha_{k+1} + \dots + c_n\alpha_n$$

Rearranging all terms to one side yields:

$$b_1\alpha_1 + \dots + b_k\alpha_k - c_{k+1}\alpha_{k+1} - \dots - c_n\alpha_n = 0$$

Since the set $\{\alpha_1, \dots, \alpha_n\}$ forms a basis for V , it is linearly independent. Therefore, the only way a linear combination of these vectors can equal the zero vector is if all the scalar coefficients are exactly zero. This implies that:

$$b_1 = \dots = b_k = 0 \quad \text{and} \quad -c_{k+1} = \dots = -c_n = 0$$

Consequently, $c_{k+1} = \dots = c_n = 0$. Because all coefficients c_i in the original equation must be zero, the set $\{T\alpha_{k+1}, \dots, T\alpha_n\}$ is linearly independent.

Rubric (10 Marks Total):

Spanning (5 Marks)

- [1 mark] Defines an arbitrary vector $w \in R(T)$ as $w = T(v)$ for some $v \in V$.

- [1 mark] Expresses v correctly as a linear combination of the basis for V : $v = \sum_{i=1}^n c_i \alpha_i$.
- [1 mark] Applies T and utilizes linearity to expand the expression into $T(v) = \sum_{i=1}^n c_i T(\alpha_i)$.
- [1 mark] Correctly identifies that $T(\alpha_i) = 0$ for $i = 1, \dots, k$ because $\alpha_i \in N(T)$.
- [1 mark] Concludes logically that w is a linear combination of $\{T\alpha_{k+1}, \dots, T\alpha_n\}$, proving it spans $R(T)$.

Linear Independence (5 Marks)

- [1 mark] Sets up the correct linear combination equated to zero: $c_{k+1}T(\alpha_{k+1}) + \dots + c_n T(\alpha_n) = 0$.
- [1 mark] Employs linearity to rewrite the equation as $T(c_{k+1}\alpha_{k+1} + \dots + c_n \alpha_n) = 0$.
- [1 mark] Logically deduces that the vector $(c_{k+1}\alpha_{k+1} + \dots + c_n \alpha_n)$ resides in the null space $N(T)$.
- [1 mark] Equates this vector to a linear combination of the null space basis: $\sum_{i=k+1}^n c_i \alpha_i = \sum_{j=1}^k b_j \alpha_j$.
- [1 mark] Rearranges the equation to equal zero and successfully invokes the linear independence of the full basis of V to conclude that all $c_i = 0$.

Q13. Let V be a vector space over $\mathbb{C}$ and suppose $T \in \mathcal{L}(V)$ is a *normal* linear operator. Using Schur's Theorem, we know that the matrix of T is upper triangular on some orthonormal basis $e_1, \dots, e_n$, i.e.

$$\mathcal{M}(T, (e_1, \dots, e_n)) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ 0 & a_{2,2} & a_{2,3} & \dots & a_{2,n} \\ 0 & 0 & a_{3,3} & \ddots & \\ 0 & 0 & 0 & \dots & a_{n,n} \end{bmatrix}.$$

Suppose $a_{1,2} = 0$. Prove that $a_{2,3} = \dots = a_{2,n} = 0$.

Answer:

Let $A = \mathcal{M}(T, (e_1, \dots, e_n))$. Since T is normal, we have

$$AA^* = A^*A.$$

A standard consequence is that the squared norm of each row of A equals the squared norm of the corresponding column.

Consider column 2. Because A is upper triangular,

$$\text{col}_2 = (a_{1,2}, a_{2,2}, 0, \dots, 0)^T,$$

so

$$\|\text{col}_2\|^2 = |a_{1,2}|^2 + |a_{2,2}|^2.$$

Given $a_{1,2} = 0$, this becomes

$$\|\text{col}_2\|^2 = |a_{2,2}|^2.$$

Now consider row 2:

$$\text{row}_2 = (0, a_{2,2}, a_{2,3}, \dots, a_{2,n}),$$

so

$$\|\text{row}_2\|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \dots + |a_{2,n}|^2.$$

Since A is normal,

$$\|\text{row}_2\|^2 = \|\text{col}_2\|^2,$$

hence

$$|a_{2,2}|^2 + |a_{2,3}|^2 + \dots + |a_{2,n}|^2 = |a_{2,2}|^2.$$

Cancelling $|a_{2,2}|^2$ from both sides:

$$|a_{2,3}|^2 + \dots + |a_{2,n}|^2 = 0.$$

Since each term is nonnegative, it follows that

$$a_{2,3} = \dots = a_{2,n} = 0.$$

Rubric:

- (a) **Uses normality correctly (2 marks):** States or uses $AA^* = A^*A$, or equivalent fact: row norms = column norms.

(b) **Computes column 2 norm (2 marks):** Identifies column 2 and shows

$$\|\text{col}_2\|^2 = |a_{1,2}|^2 + |a_{2,2}|^2,$$

then uses $a_{1,2} = 0$.

(c) **Computes row 2 norm (2 marks):** Writes

$$\|\text{row}_2\|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \cdots + |a_{2,n}|^2.$$

(d) **Equates norms and simplifies (2 marks):** Sets row norm = column norm and cancels $|a_{2,2}|^2$.

(e) **Correct conclusion (2 marks):** Concludes $a_{2,3} = \cdots = a_{2,n} = 0$ with justification (sum of squares = 0 $\Rightarrow$ each term = 0).

Q14. Let m, n be positive integers. Suppose R is an $m \times n$ row-reduced echelon matrix over some field F and denote its non-zero row vectors by $\rho_1, \rho_2, \dots, \rho_r$. Let W be its row space and suppose the vector $\beta = (b_1, b_2, \dots, b_n)$ is in W . Below, we use k_i to denote the index of the first non-zero column in the row vector ρ_i .

- i. Find the coefficients $c_1, \dots, c_r$ (in terms of $b_1, \dots, b_n$) when β is written as $\beta = c_1\rho_1 + \cdots + c_r\rho_r$.
- ii. Let $1 \leq s \leq r$. Prove that there is exactly one vector σ_s in W such that the k_s th coordinate of $\sigma_s = 1$ and the remaining k_i th coordinates (for $i \in \{1 \dots r\} \setminus s$) of σ_s are zero.
- iii. Is $\sigma_s = \rho_s$? Prove your answer.

Answer:

Part 1

To find the coefficients $c_1, c_2, \dots, c_r$ such that $\beta = c_1\rho_1 + c_2\rho_2 + \cdots + c_r\rho_r$, we can express β as a linear combination of the non-zero rows of R . Since ρ_i has a leading entry in the k_i th column, we can write:

$$\beta = c_1\rho_1 + c_2\rho_2 + \cdots + c_r\rho_r.$$

This implies that the coefficients c_i can be determined by the entries of β corresponding to the leading columns of ρ_i . Specifically, we have:

$$c_i = b_{k_i}$$

for $i = 1, 2, \dots, r$.

Part 2

To prove that there is exactly one vector σ_s such that the k_s th entry of σ_s is 1 for $i = s$ and 0 for $i \neq s$, and the remaining entries of σ_s are all zero, and $\sigma_s \in W$, we can construct such a vector σ_s as follows:

$$\sigma_s = \rho_s.$$

Since ρ_s is a non-zero row of R , it has a leading entry in the k_s th column, which is equal to 1. For all other rows ρ_i where $i \neq s$, the leading entry is in a different column, and thus the k_s th entry of σ_s will be 0. Therefore, σ_s satisfies the required conditions. Also, we can note the fact that the non-zero rows are linearly independent, since they are in Row Reduced Echelon Form. Hence, any other vector in W that has the same leading entries as σ_s must be a scalar multiple of σ_s . Since the leading entry of σ_s is 1, there can be no other vector in W that satisfies the same conditions, proving the uniqueness of σ_s .

Part 3

To determine whether $\sigma_s = \rho_s$, we can compare the entries of σ_s and ρ_s . Since σ_s is constructed to have a leading entry of 1 in the k_s th column and 0 in all other leading columns, it follows that σ_s must be equal to ρ_s . This is because ρ_s is the only non-zero row in R that has a leading entry in the k_s th column, and all other entries of ρ_s are determined by the Row Reduced Echelon Form. Therefore, we conclude that $\sigma_s = \rho_s$.

Rubric:

- **Part 1: 1 m**
 - Correctly identifies the coefficients c_i in terms of b_{k_i} : **0.5 m**
 - Provides a clear explanation of how to express β as a linear combination of the non-zero rows: **0.5 m**
- **Part 2: 3 m**
 - Mention/Show that $S = \{\rho_1, \rho_2, \dots, \rho_r\}$ is a basis for W : **0.5 m**
 - Show that there is at least one vector σ_s that satisfies the conditions: **1 m**
 - Show that any other vector in W that satisfies the same conditions must be a scalar multiple of σ_s : **1.5 m**
- **Part 3: 1 m**
 - Correctly deduces that $\sigma_s = \rho_s$: **1 m**